



Effects of beaver dams on cellulose decomposition and water quality

Azal Abbas¹ · Judit Padisák² ·
Jasmine Mancuso³ · Scott D. Tiegs³ ·
Géza B. Selmeczy²

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Abstract Beavers are keystone species that modify ecosystem structure and function through dam building, thereby creating a complex mosaic of lotic and lentic ecosystems. We assessed the influence of Eurasian beavers (*Castor fiber*) on cellulose decomposition using a cotton-strip assay and compared physical and chemical conditions between upstream reference reaches and beaver ponds in five low-order streams in

Hungary. Standardized cotton strips were deployed in both aquatic and shoreline habitats at each site. Cotton-strip decomposition, measured as tensile-strength loss, was significantly faster in upstream reference reaches than in beaver ponds and faster in aquatic habitats than in shoreline habitats. Turbidity, pH, dissolved oxygen, and oxygen saturation were higher in reference reaches than in beaver ponds, whereas most measured chemical variables did not differ significantly. Additional multivariate analysis indicated that decomposition was associated with several physical and chemical gradients rather than with any single environmental variable. Overall, our results suggest that beaver dams modify habitat conditions and reduce cellulose decomposition rates in ponded reaches. These changes may influence organic matter processing and carbon retention pathways, although direct measurements of carbon pools and greenhouse-gas fluxes are needed to evaluate whether beaver-modified habitats function as net carbon sinks.

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A. Abbas (✉)
Doctoral School of Biology, Institute of Biology, Eötvös
Loránd University, 1117 Budapest, Hungary
e-mail: azaleco@student.elte.hu

A. Abbas · J. Padisák · G. B. Selmeczy
Research Group of Limnology, Center for Natural Science,
University of Pannonia, Egyetem u. 10, 8200 Veszprém,
Hungary

A. Abbas
Department of Diwaniyah Education, Ministry
of Education, Diwaniyah City 58001, Iraq

J. Padisák · G. B. Selmeczy
HUN-REN-PE Limnoecology Research Group, Egyetem
u. 10, 8200 Veszprém, Hungary

J. Mancuso · S. D. Tiegs
Department of Biological Sciences, Oakland University,
Rochester, MI 48309, USA

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Introduction

Eurasian beavers (*Castor fiber*, Linnaeus 1758) have historically inhabited Europe and Asia, with their range extending south from the Arctic to the

Mediterranean Sea (Parker et al., 2012). After significant population decreases and subsequent reintroduction initiatives, beaver populations are re-establishing throughout most of their historical range resulting in ecological impacts on freshwater systems. Beavers are widely recognized as keystone species and quintessential ecosystem engineers capable of substantially modifying hydrological, geomorphological, and ecological processes through their engineering activities (Naiman et al., 1986; Moore, 2006; Puttock et al., 2017; Brazier et al., 2021).

Beavers build dams, which create impoundments upstream and introduce discontinuities in lotic systems (Polvi & Wohl, 2012), transforming them into complex mosaics of lotic and lentic ecosystems. By slowing flow and increasing water retention, beaver ponds function as natural filters that trap fine sediments and nutrients. This process influences nutrient cycling within ponds (Klotz, 1998) and may improve water quality downstream (Naiman et al., 1986). Beaver activity affects physical and chemical characteristics of water in both river channels and beaver ponds (Arndt & Domdei, 2011; Fracz & Chow-Fraser, 2013; Čanády et al., 2016), often resulting in the neutralization of pH, retention of nutrients and heavy metals, and reduced oxygen concentrations relative to free-flowing stream reaches (Lazar et al., 2015). Additionally, beaver ponds serve as reservoirs for both inorganic sediments, including sand, gravel, and fine particles, as well as organic material originating from the riparian zone (Butler & Malanson, 1995, 2005; Bylak & Kukuła, 2022).

Stream ecosystems play a significant role in carbon cycling, although they occupy a relatively small portion of the landscape (Lowe & Likens, 2005; Cole et al., 2007; Aufdenkampe et al., 2011). Streams are now acknowledged as dynamic biogeochemical reactors that alter, retain, and mineralize organic matter via microbial and ecosystem processes, rather than only serving as downstream transport conduits (Battin et al., 2008; Tank et al., 2010). For instance, Casas-Ruiz et al. (2017) demonstrated that river segments can operate as either passive transport conduits or active biogeochemical reactors for dissolved organic matter (DOM), with water residence time serving as a critical determinant of in-stream DOM processing. Raymond et al. (2016) proposed the Pulse–Shunt Concept, elucidating how hydrological events govern the transport and processing of dissolved organic

matter (DOM) by swiftly conveying substantial DOM pulses downstream during high discharge times. These viewpoints emphasize the role of local hydrological and biogeochemical variables in regulating carbon dynamics within stream networks.

Beaver impoundments can alter ecological functioning and biogeochemical processes, including organic matter decomposition (Catalán et al., 2017). Organic matter decomposition is a vital ecological process that contributes to the global carbon cycle, regulates atmospheric CO₂ levels (Battin et al., 2008, 2009), and impacts food-web dynamics via carbon mineralization (Gessner et al., 2010). The activity of beavers can elevate concentrations of dissolved organic carbon in beaver ponds (Law et al., 2016; Gatti et al., 2018) due to increased sediment and nutrient retention, coupled with the overall expansion of the wetland area, which fosters an environment rich in organic matter (Vecherskiy et al., 2011). Additionally, dams capture sediment-bound particulate carbon, indicating that ponds can function as net carbon reservoirs (Naiman et al., 1986; Lizarralde et al., 1996; Correll et al., 2000). The fate of organic matter in streams, such as microbial mineralisation or long-term storage, depends on decomposition rates (Tiegs et al., 2024).

Measures of ecological function, such as respiration, primary production, metabolism, and decomposition, are increasingly included in assessments of ecological integrity in streams (Bunn & Davies, 2000; Woodward et al., 2012). Importantly, organic matter decomposition is sensitive to many variables likely to be impacted by beaver activity, including temperature (Griffiths & Tiegs, 2016; Tiegs et al., 2019, 2024), nutrient loading (Piggott et al., 2015; Mancuso et al., 2023), riparian community composition (Ferreira et al., 2020), communities of microorganisms (Fell et al., 2021), and sediment size (Tiegs et al., 2013). This sensitivity makes decomposition rate a candidate metric for exploring the effects of beavers on carbon cycling.

In Central and Eastern Europe, beaver populations have expanded rapidly in recent decades, and their ecological impacts are becoming increasingly evident. In Hungary, beaver activity has been shown to influence hydrology, vegetation dynamics, and land-use interactions, highlighting both ecological benefits and emerging management challenges (Czabán & Juhász, 2024). Understanding how this

ecosystem engineering affects carbon processing is therefore critical for evaluating the broader ecological consequences of beaver re-establishment. This study explores how habitat modification by Eurasian beavers (*Castor fiber*) alters stream ecosystem characteristics and influences carbon cycling through organic matter decomposition in five streams in Hungary. We do this by addressing the following hypothesis:

Hypothesis 1 Beaver-modified reaches alter the physical and chemical characteristics of stream ecosystems, resulting in reduced flow velocity and dissolved oxygen, alongside increased water retention and nutrient concentrations compared to unmodified reference sites.

Hypothesis 2 Beaver-induced alterations to hydrological and chemical conditions affect ecosystem

functioning, leading to reduced organic matter decomposition rates in beaver-modified reaches relative to flowing reference sections.

Materials and methods

Study area

We chose five streams [Rétkerti-patak: RP, Gyimóti-séd: GYS, Mezőlaki-séd: MLS, Ország-patak: PER, Bánya-ér: BAE (Supplementary Figs. 1, 2)] in Hungary (Fig. 1) with significant beaver activity including dam maintenance and modification of the area. The WWF Hungary released 234 individuals as part of a beaver reintroduction program during 1996–2008 (Bajomi, 2011). The program has been successful, with the population growing to 10,000–11,000 individuals by 2020 (Czabán & Juhász, 2024). The

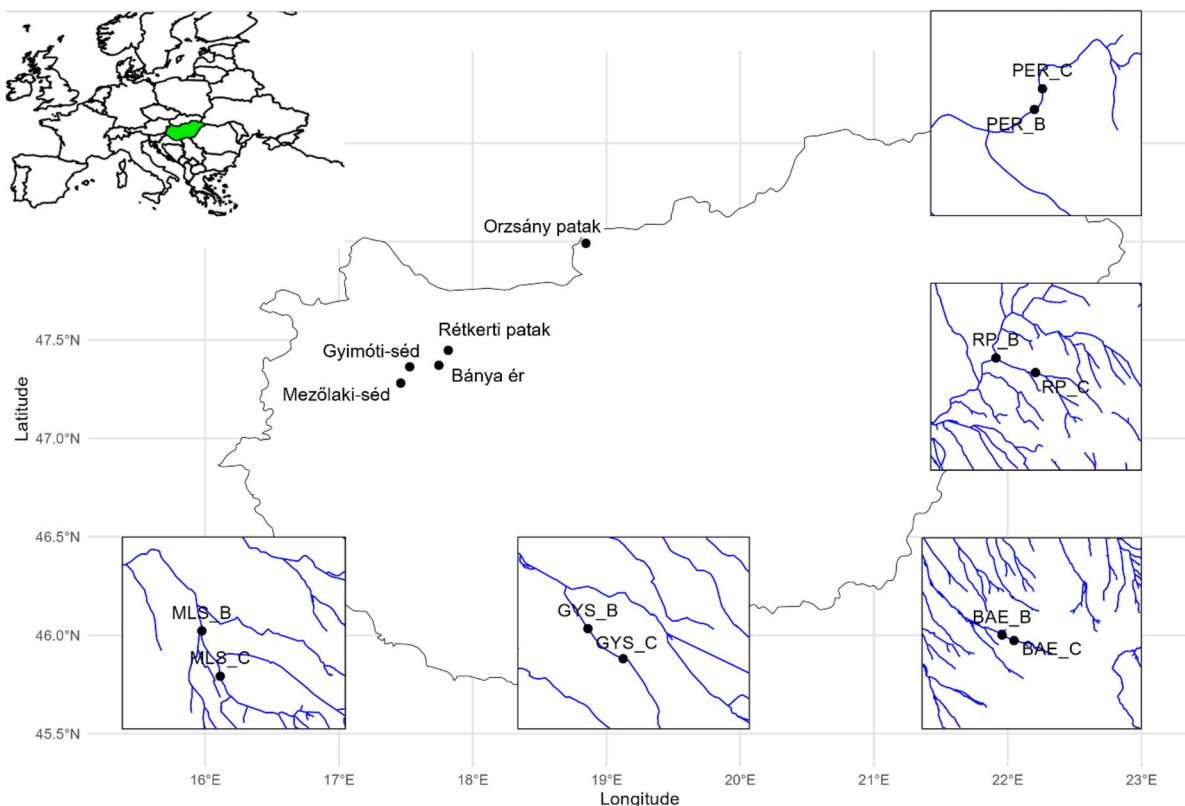


Fig. 1 Location of sampling area. Each dot represents a beaver dam impoundment and nearby reference site, and paired water-shoreline habitats. The zoomed-in maps show the location of control sites (C) and beaver ponds (B). The short names of the

stream are the followings: Mezőlaki-séd (MLS), Gyimóti-séd (GYS), Bánya-ér (BAE), Rétkerti patak (RP), Ország-patak (PER)

beaver dams were at least two years old and were constructed on small streams where the bed material is characterized by pebble, gravel, and sand. Each stream had two sites: an upstream reference or control site (C) and a beaver pond (B). The mean distance between the pairs of B and C sites was 1273 m (Supplementary Table 1). The streams at the reference sites were typically 1–2 m wide and the average depth of the water during baseflow ranged between 5 and 25 cm. The beaver ponds occupied a 30–60 m long section of the stream, extending between the dam itself and the point upstream where the flow velocity was $< 0.01 \text{ m sec}^{-1}$. The maximum depth of the beaver ponds ranged from 80 to 170 cm immediately upstream of the dam. The length of the dams ranged from 5 to 23 m. Each C and B site contained a shoreline (S) and a water (W) habitat.

Measurement of environmental variables

We measured water temperature, pH, dissolved oxygen (DO), oxygen saturation (DO%), conductivity, and turbidity weekly in situ (5 measurements total) during the experiment using a Hach Lange HQ40d multimeter and a Hach Lange TS portable turbidimeter. We collected water samples at each site twice at the beginning and end of the experiment and determined: total phosphorus (TP), soluble reactive phosphorus (SRP), ammonium (NH_4^+), nitrite (NO_2^-), soluble reactive silica (SRSi), chloride (Cl^-), bicarbonate (HCO_3^-), sulfate (SO_4^{2-}), and chemical oxygen

demand (COD) according to the APHA (2012) standards. A Hach Lange ISENO3181 nitrate probe was used to measure the nitrate (NO_3^-) concentration. We recorded continuous water temperature and air temperature every 10 min in the W habitat in the C and B reaches of streams BAE and GYS and in the S habitat in the C and B reaches of BAE during the experiment using data loggers (HOBO Pendant MX Temperature/Light Data Logger, LI-COR, Lincoln, Nebraska, USA). Data loggers were attached to the same poles used for affixing cotton strips.

Cotton-strip preparation and field experiments

To assess cellulose decomposition rates at each site (C and B) and habitat type (S and W), we used the cotton-strip assay, which quantifies tensile strength loss of cotton fabric during incubation in the field, a process that corresponds to the microbial catabolism of cellulose (Tiegs et al., 2013, 2024). At each site, four strips were deployed on the shoreline (S) and in the water (W) (Fig. 2). Altogether, we used 80 strips ($5 \text{ streams} \times 2 \text{ sites} \times 2 \text{ habitats} \times 4 \text{ strips}$). We made the cotton strips from a single bolt of Fedrix-brand 12-oz., unprimed, heavy-weight cotton fabric (Lawrenceville, GA, USA; style #548). We conducted the experiment between the 2nd and 30th of October 2023 (incubation period = 29 days). The assay was applied according to the standard methods described by Tiegs et al. (2013, 2024) and as used by others (e.g., Mancuso et al., 2023). A pointed and slender

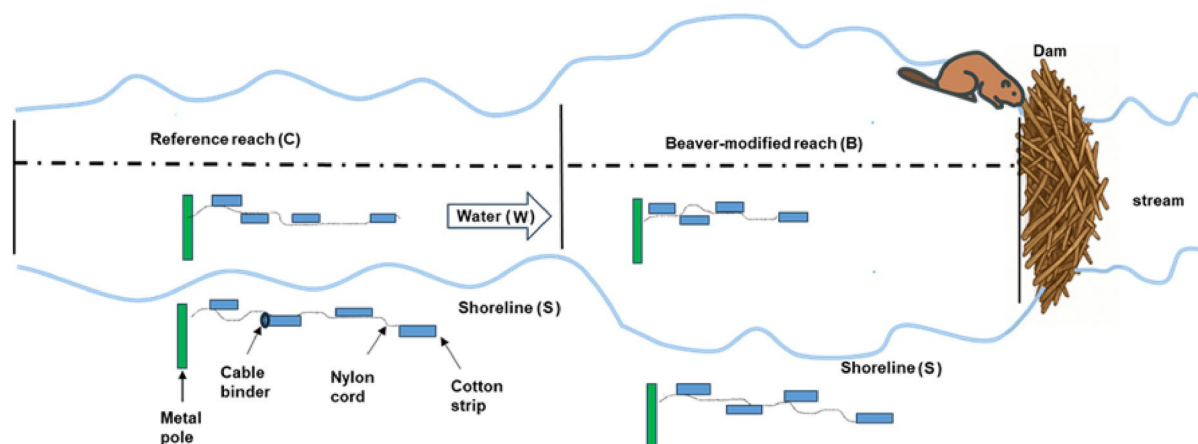


Fig. 2 A schematic showing the relative position of the reference (C) and beaver-modified reaches (B) of a study stream with cotton-strip deployments at water (W) and shoreline (S) habitats

tool was used to create a small space between the threads near one end of each cotton strip, and a thin cable binder was inserted through the hole to bind the four strips to a nylon cord 10–15 cm long. We attached the cord to two metal poles at each site, one of which was fixed in the streambed and the other in the soil on the shoreline (Fig. 2). After the 29-day incubation period, the cotton strips were collected and gently cleaned with 70% ethanol for a few seconds to halt microbial activity. Each strip was placed individually in a labeled aluminum foil envelope. Two cotton strips (one from RP-C-W and the one from BAE-C-W) were fully decomposed and carried away by the current (Supplementary Table 2). In the lab, the cotton strips were dried at 40 °C for 24 h. The strips were shipped to Oakland University (Rochester, Michigan, USA) for tensile strength measurement.

Following Tiegs et al. (2013), we measured the remaining tensile strength of each strip using a tensiometer (Mark-10 brand, Model # MG100, Copiague, NY, USA) and pulled them at a rate of 20 mm/min. Additionally, we determined the tensile strength of 10 control strips (those not incubated in the field) to estimate initial tensile strength value prior to incubation in the field. We used the following equation to calculate the decomposition rate (as a standardized proxy of decomposition, k_d (day⁻¹)) for each strip incubated in the field:

$$k_d = (-\ln(T_s/T_{sc}))/\text{days}$$

where T_s represents the maximum tensile strength recorded for each incubated strip, T_{sc} is the mean tensile strength of the control strips (i.e., the average tensile strength of 10 strips that were not incubated in the field), and days refers to the amount of time the cotton strips were incubated in the field.

Statistical analysis

Data analyses were carried out in R 4.5.1 environment (R Core Team, 2025). Linear mixed model was used to explore the changes of tensile strength of cotton strips (as k_d values) with the *lmer* function in the “lme4” package (Bates et al., 2015). Site (B vs. C), Habitat (S vs. W), and their interaction were included as fixed effects, while Stream (five streams) was treated as a random intercept. Distribution of the residuals was checked with *qqnorm* and

qqline functions. Models were fitted using restricted maximum likelihood (REML). Statistical significance of fixed effects was assessed using Satterthwaite’s approximation for degrees of freedom, using “lmerTest” package (Kuznetsova et al., 2017). Linear mixed models were used to investigate both the relationship between k_d values and individual environmental variables and the combined effects of environmental conditions. However, prior to the analysis, principal component analysis (PCA) was performed on standardized environmental variables including temperature, DO, pH, conductivity, turbidity, TP, COD, NO_3^- , and NH_4^+ using the *prcomp* function in the “vegan” package (Oksanen et al., 2022). The PCA component scores were used as explanatory variables in the model. Visualization of the data was carried out with “ggplot2” (Wickham, 2016), “ggsignif” package (Constantin & Patil, 2021), “sf” package (Pebesma & Bivand, 2023), “osmdata” package (Padgham et al., 2017), and “factoextra” package (Kassambara & Mundt, 2026). Prior to statistical analyses, normality of physical and chemical variables was examined graphically using histogram. A paired *t* test was conducted to examine the differences in physical and chemical variables pH, dissolved oxygen (DO), oxygen saturation, temperature, conductivity, HCO_3^- , SO_4^{2-} , chemical oxygen demand (COD), and soluble reactive silica (SRSi) between the reference reaches and the beaver ponds, assuming normal distribution. For variables that failed the normality test, such as turbidity, total phosphorus (TP), soluble reactive phosphorus (SRP), nitrate (NO_3^-), nitrite (NO_2^-), ammonium (NH_4^+), and chloride (Cl^-), a non-parametric Wilcoxon test was employed. A significant difference was specified at $P < 0.05$, and a highly significant difference at $P = 0.0001$ and $P < 0.0001$. Statistical analyses of physiochemical variables were conducted using the GraphPad Prism 8.

Results

Physical and water chemical variables

Values of pH, DO, oxygen saturation, and turbidity (Supplementary Table 4) were significantly higher at the reference reaches than in the beaver ponds (Table 1 and Supplementary Fig. 4a–d). However, there were no significant differences in water

Table 1 Physical and chemical variables measured in reference reaches and beaver ponds

Physical and chemical variables of the water	Reference reaches (C) Mean $\pm$ Sd	Beaver ponds (B) Mean $\pm$ Sd	T test (T)	Wilcoxon test	P values	Sample size (n)
pH	8.0 $\pm$ 0.2439	7.8 $\pm$ 0.1778	t = 4.998, df = 20	–	< 0.0001	21
DO (mg l ⁻¹)	6.173 $\pm$ 2.372	4.394 $\pm$ 2.848	t = 4.817, df = 20	–	0.0001	21
Oxygen saturation (%)	62.59 $\pm$ 22.02	42.70 $\pm$ 28.09	t = 5.302, df = 20	–	0.0001	21
Turbidity (NTU)	16.25 $\pm$ 13.09	8.75 $\pm$ 7.17	–	– 127	0.0263	21
Water temperature (°C)	15.2 $\pm$ 1.74	14.7 $\pm$ 1.4	t = 1.315, df = 20	–	0.2032	21
Conductivity (μ S cm ⁻¹)	860.0 $\pm$ 255.2	861.5 $\pm$ 246	t = 0.1046, df = 20	–	0.9177	21
HCO ₃ ⁻ (mg l ⁻¹)	5.2 $\pm$ 1.697	5.6 $\pm$ 2.03	t = 2.236, df = 9	–	0.0522	20
TP (mg l ⁻¹)	0.7 $\pm$ 1.19	0.3 $\pm$ 0.55	–	– 19	0.3008	20
SRP (mg l ⁻¹)	0.3 $\pm$ 0.61	0.2 $\pm$ 0.36	–	– 31	0.0742	20
NO ₃ ⁻ (μ g l ⁻¹)	4.7 $\pm$ 4.83	3.5 $\pm$ 4.06	–	– 25	0.2324	20
NO ₂ ⁻ (mg l ⁻¹)	0.05 $\pm$ 0.07	0.03 $\pm$ 0.02	–	– 7	0.7695	20
NH ₄ ⁺ (mg l ⁻¹)	0.2 $\pm$ 0.31	0.1 $\pm$ 0.06	–	1	0.9805	20
SO ₄ ²⁻ (mg l ⁻¹)	46.4 $\pm$ 34.47	52.3 $\pm$ 42.75	t = 1.803, df = 9	–	0.1049	20
COD (mg l ⁻¹)	2.90 $\pm$ 2.21	3.41 $\pm$ 1.89	t = 0.8559, df = 9	–	0.4143	20
SRSi (mg l ⁻¹)	6.9 $\pm$ 2.97	7.1 $\pm$ 3.54	t = 0.5383, df = 8	–	0.6050	19
Cl ⁻ (mg l ⁻¹)	41.83 $\pm$ 52.38	31.00 $\pm$ 26.73	–	11.00	0.5703	20

All values are reported as mean $\pm$ standard deviation (SD), with sample size indicated as (n). Statistically significant results are indicated in bold. Abbreviations are defined as follows: DO dissolved oxygen, HCO₃⁻ bicarbonate, TP total phosphorus, SRP soluble reactive phosphorus, NO₃⁻ nitrate, NO₂⁻ nitrite, NH₄⁺ ammonium, SO₄²⁻ sulfate, COD chemical oxygen demand, SRSi soluble reactive silica, and Cl⁻ chloride

temperature and conductivity between the references and beaver-modified reaches (Table 1 and Supplementary Fig. 4e–f).

Water chemistry variables (HCO₃⁻, TP, SRP, NO₃⁻, NO₂⁻, NH₄⁺, SO₄²⁻, COD, SRSi, and Cl⁻) (Supplementary Table 5) showed no significant differences between the reference reaches and the beaver ponds (Supplementary Figs. 4a–f and 5a–d). The standard deviation for TP was large (Table 1). Both TP and SRP were lower, whereas SO₄²⁻ was higher, in the beaver ponds than at the reference reaches. However, the differences were not significant statistically (Supplementary Figs. 4b, c and 5a).

Temperature

The air temperature reached a minimum value of 0.9 and a maximum value of 21.5 °C at the shoreline of the beaver pond in the BAE, while at the reference reach it ranged between 4.5 and 20.8 °C (Supplementary Fig. 3a). At the same location, water temperatures ranged from 6.0 to 18.3 °C in the beaver pond

and 6.7 to 15.7 °C at the reference reach (Supplementary Fig. 6b). At the GYS location, water temperatures varied between 10.4 and 17.2 °C at the beaver pond, while they ranged from 8.6 and 16.6 °C at the reference reach (Supplementary Fig. 6c).

Cellulose decomposition

The tensile strength loss (as a standardized proxy of decomposition k_d) was significantly higher at the reference reaches (0.070 $\pm$ 0.055), than in the beaver ponds (0.031 $\pm$ 0.029) in the aquatic habitat (W) (Fig. 3). However, no differences were observed in k_d between the reference reaches (0.007 $\pm$ 0.003) and the beaver ponds (0.006 $\pm$ 0.003) at the shoreline (S) according to the results of the linear mixed-effect model (Supplementary Table 3, Fig. 3).

Individual environmental variables were first tested with linear mixed-effect models and, individually, none of the variables significantly affected decomposition, including DO (estimate = - 0.011 $\pm$ 0.007 SE, t = -1.71, P = 0.125). However, PCA (Fig. 4) was used to summarize variation in environmental conditions, and a linear mixed-effect model

Fig. 3 Boxplots representing decomposition rates (k_d) of cotton strips, measured as tensile strength loss of the cotton fabric over the incubation time, from the two site types (beaver ponds and reference reaches) and habitat types (shoreline and water) ($n = 78$). Significant differences are denoted by an asterisk * ($P < 0.05$), *** ($P = 0.0001$) and **** ($P < 0.0001$). No significant difference is labeled as “NS.” Shortenings are the following: C (reference reach), B (beaver pond), S (shoreline), and W (water)

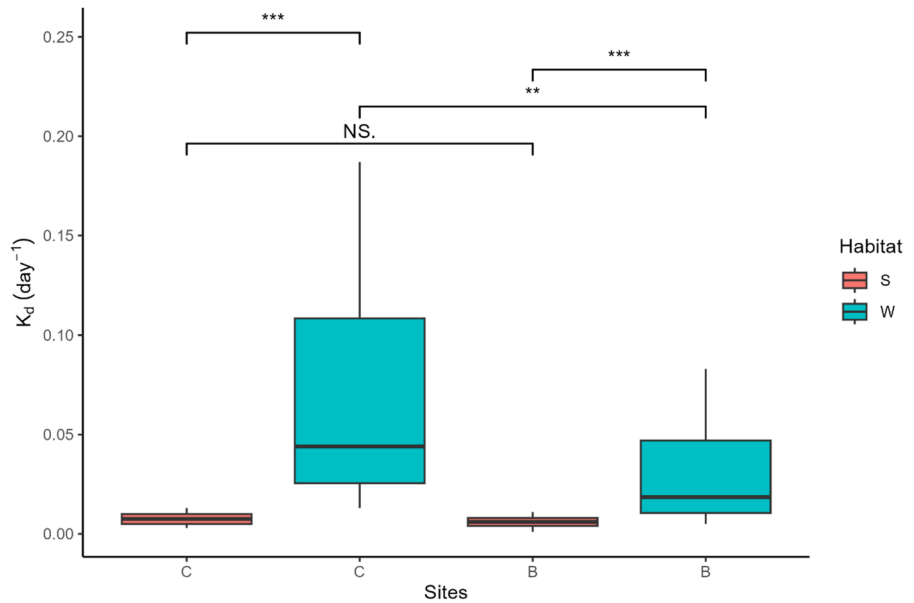
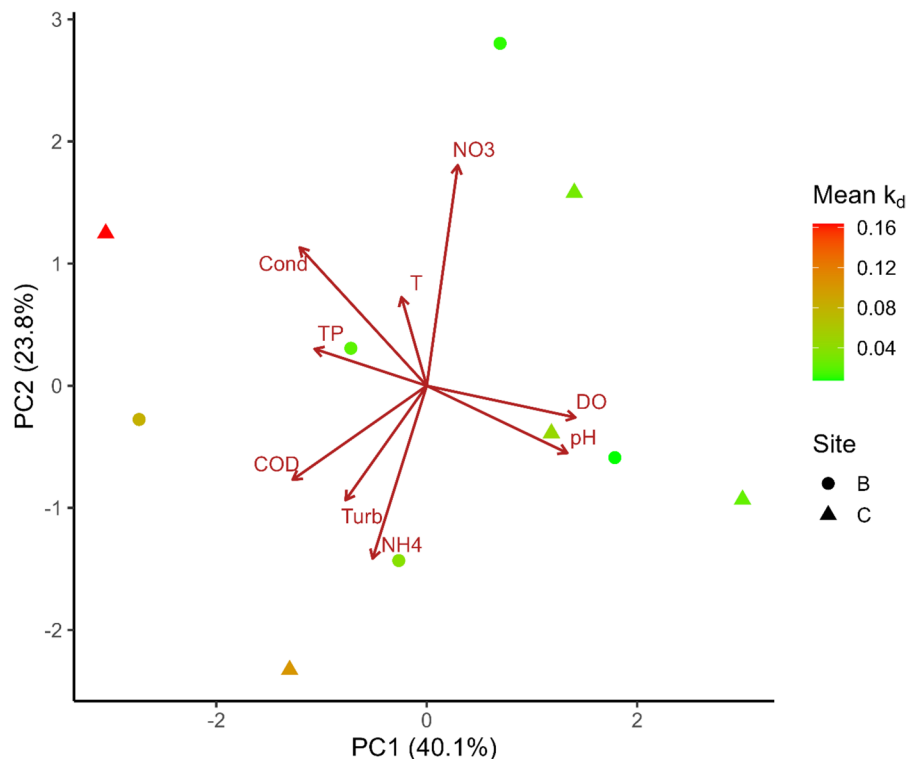


Fig. 4 Principal component analysis biplot based on the decomposition rate (k_d) and the most important environmental variables. T water temperature, NO_3 nitrate, DO dissolved oxygen, NH_4 ammonium, $Turb$ turbidity, COD chemical oxygen demand, TP total phosphorus, $Cond$ conductivity



found a significantly negative relationship between decomposition (k_d) and the first principal component (estimate = -0.0189 ± 0.0031 SE, $P = 0.011$), whereas PC2 had no significant effect ($P = 0.311$).

The first two principal components explained 40.1% and 23.8% of the total environmental variation, respectively. PC1 was primarily associated with DO , pH , COD , and conductivity, while PC2 was

mainly linked with NO_3^- , conductivity, temperature, and NH_4^+ (Fig. 4 and Supplementary Table 6). It is important to note that the effective degrees of freedom associated with the PC1 were relatively low ($df = 2.75$).

Discussion

We assessed cellulose decomposition rates in beaver ponds (B) and upstream reference reaches (C) using the cotton-strip assay to evaluate how beaver dams influence microbial decomposition processes, an important component of organic matter processing in-stream ecosystems. Additionally, we compared physical and chemical variables between reference reaches and beaver-modified habitats across five streams to explore potential environmental drivers of decomposition. The findings indicated that beaver activity significantly changed the decomposition of organic matter, leading to lower decomposition rates in the beaver ponds compared to the upstream reference reaches. This reduction was accompanied by significant shifts in several physical and chemical variables, including lower pH, dissolved oxygen, oxygen saturation, and turbidity. Although no individual environmental variable significantly explained decomposition rates, multivariate analysis indicated that decomposition was associated with the primary environmental gradient across habitats. Together, these findings suggest that beaver impoundments influence decomposition through integrated habitat-scale changes rather than through the effect of a single physical or chemical factor. However, given the limited number of independent streams–habitat combinations, these relationships should be interpreted cautiously.

Water physical and chemical variables

Levels of pH were slightly alkaline in our study, consistent with the general characteristics of many freshwater ecosystems in this region (Borics et al., 2003; Ishaq & Khan, 2013). However, significantly lower values of pH were observed at the beaver ponds compared to upstream reference reaches. This reduction in pH may reflect physicochemical changes associated with beaver impoundment, including increased water residence time, enhanced retention of organic matter, and altered microbial processing (Larsen

et al., 2021). Previous studies have shown that the accumulation and decomposition of organic material in beaver ponds can increase the production of dissolved organic compounds and organic acids, which may contribute to lower pH (David & Vance, 1991; Cirimo & Driscoll, 1993). Thus, the lower pH observed in beaver ponds in our study is consistent with enhanced organic matter retention and processing within impounded habitats.

In agreement with previous studies (Otyukova, 2009; Levanoni et al., 2015; Vehkaoja et al., 2015), DO concentrations and oxygen saturation were significantly lower in the beaver ponds than in the reference reaches. Although mean oxygen saturation did not indicate persistent anoxic conditions, occasional very low DO values were recorded (minimum = 0.61 mg l^{-1} ; Supplementary Table 4), indicating periods of substantially reduced oxygen availability. These lower oxygen levels may reflect hydrological changes associated with impoundment, including reduced flow velocity, increased water residence time, and greater retention of organic matter (Pollock et al., 2003; Green & Westbrook, 2009). Previous studies suggest that the accumulation and microbial processing of retained organic material, including woody debris transported by beavers (Ecke et al., 2017; Rozhkova-Timina et al., 2018), can contribute to increased biological oxygen demand in ponded habitats. However, in our study, dissolved oxygen alone did not significantly explain decomposition rates, suggesting that oxygen differences should be interpreted as one component of a broader physicochemical gradient rather than as an isolated driver of decomposition. Although nutrient concentrations did not differ significantly between habitats, the combined effects of hydrological alteration, organic matter retention, and physicochemical changes may collectively influence ecosystem functioning in beaver-modified reaches.

The reduction in water velocity in beaver ponds, which is associated with a decrease in the water surface gradient (Giri et al., 2016) and consequently stream power (Butler & Malanson, 1995), results in increased sedimentation of suspended particles. As reported by Puttock et al. (2018), beaver ponds have been shown to function as effective sediment sinks, storing large amounts of mineral and organic material and thereby reducing turbidity in the water column (Larsen et al., 2021). Beaver dams therefore function as sediment traps, capturing finer particles

over time (Pollock et al., 2007), which can reduce turbidity within the pond itself as reported by White et al. (2025). This interpretation is consistent with our findings, where turbidity was significantly lower in the beaver ponds, whereas higher turbidity and higher decomposition rates were recorded in the reference reaches, in line with the positive relationship between turbidity and decomposition reported by Mancuso et al. (2023).

Temperature is recognized as a “master driver” that determines the potential maximum rate of decomposition in streams, as microbial activity is positively affected by temperature, except under extreme conditions (Suberkropp & Chauvet, 1995; Follstad Shah et al., 2017; Tiegs et al., 2019; Costello et al., 2022). However, our findings revealed that the water temperature in beaver ponds did not significantly differ from the reference reaches. Field observations indicated that the maximum depth of the beaver ponds ranged from 80 to 170 cm immediately upstream of the dam and shading from surrounding vegetation. According to McRae and Edwards (1994), the short retention time and shallow depth of beaver ponds lead to the absence of thermal stratification, without a division into a lower or upper outlet in the dam. As a result, water flows across the entire surface of the dam, which can help buffer thermal variability and stabilize stream temperature. Our in situ temperature data at BAE and GYS (Supplementary Fig. 3) collected by datalogger revealed slight, but insignificant, elevations in temperature at the beaver ponds compared to reference reaches. Although certain studies have indicated that beaver ponds are warmer than adjacent stream reaches (Naiman et al., 1994; Hartman & Törnlov, 2006; Błędzki et al., 2011), the thermal responses to beaver engineering are highly context dependent. The absence of temperature differences in our study is likely due to the intense riparian shading and hydrological buffering.

Conductivity is an important environmental variable that influences both planktonic (Padisák & Naselli-Flores, 2021) and benthic (Stenger-Kovács et al., 2023) primary producers. Elevated ionic content can enhance algal productivity and thus influence the quantity of organic matter. However, in agreement with previous studies (Hood & Larson, 2014; Vehkaoja et al., 2015; Gatti et al., 2018), our study showed no significant difference in conductivity values

between the reference reaches and beaver ponds, suggesting that ionic conditions were comparable across the studied streams.

Water chemistry can be influenced by beaver activity, with effects often becoming more pronounced as ponds mature and hydrological residence time increases (Collen & Gibson, 2000; Błędzki et al., 2011; Law et al., 2017; Puttock et al., 2017). However, none of the measured chemical variables (HCO_3^- , TP, SRP, NO_3^- , NO_2^- , NH_4^+ , COD, SO_4^{2-} , Cl^- , and SRSi) differed significantly between reference reaches and beaver ponds in this study. This limited differentiation may be partially due to the comparatively young age (as noted in this study) or hydrological connectivity of the studied ponds, as older beaver impoundments typically exhibit stronger biogeochemical modification (Margolis et al., 2001; Hood & Larson, 2014).

Although no significant chemical differences were detected, some consistent trends were observed. Phosphorus concentrations (TP and SRP) tended to be lower in beaver ponds, which may indicate localized retention through sedimentation under reduced flow conditions (Maret et al., 1987; Dillon et al., 1991; Puttock et al., 2017). In contrast, inorganic nitrogen species and other solutes, including sulfate, COD, SRSi, and chloride, showed little variation between habitats. Slightly elevated sulfate concentrations in ponds may be associated with enhanced organic matter processing and redox-related transformations (although such mechanisms were not directly measured here), processes that are typically associated with low-flow beaver wetlands as reported in previous studies (Naiman et al., 1994; Hood & Larson, 2014; Waş et al., 2025). The results indicate that chemical differences between habitats were minimal and the observed variation in decomposition was more likely linked to extensive habitat-scale physicochemical gradients rather than to any individual water chemistry variable.

Cellulose decomposition

Organic matter decomposition is an essential ecosystem process in aquatic environments. In this study, we assessed cellulose decomposition using the cotton-strip assay, a standardized method sensitive to multiple environmental conditions influencing microbial decomposition (Gessner & Chauvet, 2002; Tiegs et al., 2013, 2024), including those impacted by beavers.

These conditions include stream flow, which can influence organic matter decomposition directly through physical abrasion and changes of oxygen levels and indirectly by stimulating microbial activity via delivery of limiting nutrients at higher rates than passive diffusion alone (Gessner et al., 2010; Arroita et al., 2015). We observed significantly lower decomposition rates in beaver ponds compared with upstream reference reaches, indicating that beaver impoundments altered habitat conditions relevant to decomposition processes. Although the cotton strips remained intact during incubation, suggesting that direct physical abrasion was limited, differences in water velocity between flowing and ponded habitats may still have contributed to the observed decomposition patterns. Nevertheless, the reduced decomposition in beaver ponds was more likely driven by broader habitat-scale changes associated with impoundment, including increased water residence time, greater organic matter retention, and shifts in physicochemical conditions (Naiman et al., 1994; Pollock et al., 2014; Brazier et al., 2021; Larsen et al., 2021). While several environmental variables, including dissolved oxygen and pH, differed between habitats, none individually showed a significant relationship with decomposition rate. In contrast, principal component analysis revealed a significant association between decomposition and the primary environmental gradient (PC1), which explained 40.1% of the total environmental variation and was mainly characterized by variation in dissolved oxygen, pH, conductivity, and COD. These findings suggest that decomposition was influenced by the combined effects of multiple interacting environmental factors rather than by any single physical or chemical variable, consistent with previous studies demonstrating multivariate controls on cellulose decomposition in freshwater ecosystems (Gessner et al., 2010; Arroita et al., 2015; Tiegs et al., 2024). Such multivariate control is consistent with the complex habitat modifications induced by beaver dams, which simultaneously affect hydrology, sediment retention, organic matter accumulation, and microbial habitat conditions. However, this relationship should be interpreted cautiously because the effective degrees of freedom were low ($df = 2.75$), reflecting the limited number of independent streams–habitat combinations.

Although the cotton-strip assay does not directly quantify carbon pools or greenhouse-gas fluxes, decomposition rate represents an important functional

indicator of organic matter processing in-stream ecosystems (Tiegs et al., 2013, 2024). The reduced decomposition rates observed in beaver ponds suggest slower microbial processing of labile organic substrates, potentially increasing organic matter residence time and enhancing retention of carbon-rich material in pond sediments and wetland soils. This interpretation is consistent with previous studies showing that beaver dams promote sediment and organic matter retention and may enhance carbon storage in beaver-modified stream corridors (Naiman et al., 1986, 1994; Wohl, 2013; Larsen et al., 2021). However, because dissolved and particulate carbon pools, sediment carbon, and greenhouse-gas fluxes were not measured, these findings should be interpreted as evidence that beaver ponds influence a key process linked to carbon cycling rather than as direct quantification of carbon storage or emissions. In our study, decomposition rates in shoreline habitats were lower than in aquatic habitats at both reference reaches and beaver ponds. Because riparian sites were exposed to broadly similar climatic conditions, these shoreline comparisons are interpreted as complementary observations rather than primary evidence of beaver effects on decomposition.

Carbon cycling and management perspective

Beaver activity plays a critical role in altering carbon dynamics through modifying decomposition rate by sequestering organic matter in lake sediments and wetland soils (Wohl, 2013; Gatti et al., 2018). In this study, the cotton-strip assay does not directly quantify carbon pools, carbon burial, or greenhouse-gas fluxes; therefore, the present study cannot directly assess the overall carbon budget of beaver-modified systems. Nevertheless, decomposition rate represents an important functional indicator of organic matter processing and can provide insight into processes linked to carbon cycling. The reduced decomposition rates observed in beaver ponds (Fig. 3) suggest that beaver impoundments may slow microbial processing of labile organic matter, potentially increase organic matter residence time and promote greater retention of carbon-rich material within pond sediments and wetland soils. This interpretation is consistent with previous studies showing that beaver dams enhance sediment accumulation, organic matter retention, and long-term carbon storage in floodplain

and wetland environments (Wohl, 2013; Johnston, 2017; Gatti et al., 2018). However, the role of beaver ponds in regional carbon budgets remains complex. While beaver impoundments may promote carbon burial, they may also increase emissions of greenhouse gases such as CO_2 and CH_4 under certain environmental conditions, potentially offsetting some sequestration benefits (Nummi et al., 2018; Castro et al., 2026). From a management perspective, these findings suggest that beaver-driven restoration may provide ecosystem services extending beyond hydrological regulation, including potential contributions to carbon retention and climate adaptation.

Overall, our results provided partial support for the first hypothesis (H1), which predicted that beaver-modified reaches alter the physical and chemical characteristics of stream ecosystems by reducing flow velocity and dissolved oxygen while increasing water retention and nutrient concentrations, which was partially supported. Beaver ponds showed significantly lower pH, dissolved oxygen, oxygen saturation, and turbidity compared with upstream reference reaches, consistent with hydrological modification by beaver activity. However, contrary to our expectation, nutrient concentrations did not differ significantly between habitats. The second hypothesis (H2), which posits that beaver-induced hydrological and chemical changes reduce organic matter decomposition rates, was demonstrated by the substantially lower cotton-strip decomposition rates observed in beaver ponds. However, no single environmental variable accounted for decomposition. Instead, decomposition was significantly correlated with a number of physical and chemical variables as was identified through principal component analysis. This suggests that the functioning of ecosystems in beaver-modified reaches is influenced by the combined effects of multiple interacting environmental factors, rather than by individual variables alone.

The study is limited by the absence of direct measurements of redox conditions, organic matter features (dissolved or particulate), and nutrient transformation mechanisms, which constrained our capacity to correlate decomposition patterns with changes in water chemistry. In addition, the limited number of independent streams–habitat

combinations reduced statistical power and requires cautious interpretation of the multivariate analyses.

Conclusion

Beaver activity can disturb the river continuum by creating ponds that significantly alter local environmental conditions, particularly pH, dissolved oxygen, oxygen saturation, and turbidity. However, there were no significant differences between upstream reference reaches and beaver ponds in the case of several ions (HCO_3^- , TP, SRP, NO_3^- , NO_2^- , NH_4^+ , SO_4^{2-} , SRSi, and Cl^-) in our study. Beaver impoundments also significantly reduced cotton-strip decomposition rates relative to upstream reference reaches, indicating altered organic matter processing in ponded habitats. The research suggests complex habitats created by beavers may have potential enhance microbial diversity, but it would be important to confirm it with molecular studies. However, multivariate analysis further showed that decomposition was associated with a broader physicochemical gradient rather than with any single environmental variable, suggesting that beaver dams influence ecosystem functioning through integrated habitat-scale changes. Overall, our results indicate that beaver dams modify habitat conditions and decomposition dynamics within stream ecosystems. While these changes may influence carbon-processing pathways, their implications for carbon storage, sequestration, and ecosystem-scale carbon budgets require further investigation through direct measurements.

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Data availability Original dataset is uploaded as supplementary file.

Declarations

Conflict of interest The authors declare that they have no conflict of interest apart that Judit Padisák is an Associate Editor

of the Hydrobiologia therefore must be blinded during processing this manuscript.

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